MIL-C-63405 (AR)
11 September 1980

MILITARY SPECIFICATION
CHARGE, PROPELLING, 155MM, M119A2
LOADING, ASSEMBLING, AND PACKING

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, 155MM: M119A2, for 155MM Howitzers, M109A1/A2/A3 and M198.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

DRAWINGS

ARRADCOM

PRODUCT AND PACKAGE DRAWINGS

9333954	- Charge, Propelling, 155MM: M119A2
9333957	- Packing and Marking for Container, Ammo, Metal: PA37A1 for Charge, Propelling, 155MM, M119A2

☆U.S. GOVERNMENT PRINTING OFFICE: 1980—703-023/4313 FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials - Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling Charge - The propelling charge shall comply with all requirements specified on drawing 9333954 and with all requirements specified in applicable specifications.

3.3 Moisture Content of Black Powder - The moisture content of the black powder at the igniter loading station at the time of loading shall not exceed the amount specified in MIL-P-223.

3.4 Weights

3.4.1 Propellant Weight - The propellant weight shall comply with the weight specified in the loading authorization within plus or minus 0.20 ounce at the time of loading.

3.5 Dimensional Gage

3.5.1 Maximum Length and Diameter - The completed charge shall freely enter the prescribed gage or gages. If the diameter feature is separate from the length feature, the diameter feature must gage the full length of the charge.

3.6 Proving Ground (see 6.7) - The muzzle velocity, pressure and velocity standard deviation shall be as specified in Table 1.

TABLE 1

Zone	Proj	Temp	Velocity (fps) Minimum	Velocity (fps) Maximum	Velocity Standard Deviation FPS	Individual Maximum Pressure PSI
7	107	70°F	2226	2276	10.0	--
7	107	145°F	—	—	—	41,000

3.7 Assembly Torque - The test plug shall be assembled to withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving (see 6.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.8 Charge Assembly Height - An approved method of measuring the vertical height of the loaded charge assembly (dwg 9340443) will be utilized for all charges.

3.9 First Article Inspection - This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship - All components shall be free of dirt, grease and other foreign material. There shall be no discoloration or deterioration of cloth and no evidence of rips, tears, or other cloth defects. All work shall be performed in a thorough workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. This sample shall consist of the following:

Part Description	Drawing	Quantity
Strap	9340442	5
End Spot	9340455	5
End, Charge	9340450	5
Laminated Rayon Cloth	9340449	5
Body	9340448	5
End, Base Igniter	9340456	5
Spot Assembly	9340454	5
End & Spot Assembly	9340453	5
Base Igniter Assembly	9340452	5
Base Igniter Loading Assembly	9340451	5
Body & Liner Assembly	9340447	5
Flash Reducer Loading Assembly	9340446	5
End Charge and Flash Reducer Loading Assembly	9340445	5
End Charge and Flash Reducer Loading Assembly and Base Igniter Loading Assembly	9340444	5
Charge Loading Assembly	9340443	5
Propelling Charge, 155MM, M119A2 (in sealed containers)	9333954	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.3.2 Inspections to be Performed - The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table II. All inspections listed in Table II shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3 Rejection - See MIL-A-48078

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Propelling Charge Components and Assemblies				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Strap (Dwg 9340442) Examination for defects	5		3.2	4.4.2.1
	End, Spot (Dwg 9340455) Examination for defects	5		3.2	4.4.2.2
	End, Charge (Dwg 9340450) Examination for defects	5		3.2	4.4.2.3
	Laminated Rayon Cloth (Dwg 9340449) Examination for defects	5		3.2	4.4.2.4
	Body (Dwg 9340448) Examination for defects	5		3.2	4.4.2.5
	End, Base Igniter (Dwg 9340456) Examination for defects	5		3.2	4.4.2.6

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2025-10-26 04:27Z
Check the source to verify that this is the current version before use.

TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spot Assembly (Dwg 9340454) Examination for defects	5		3.2	4.4.2.7
	End & Spot Assembly (Dwg 9340453) Examination for defects	5		3.2	4.4.2.8
	Base Igniter Assembly (Dwg 9340452) Examination for defects	5		3.2	4.4.2.9
	Base Igniter Loading Assembly (Dwg 9340451) Examination for defects	5		3.2	4.4.2.10
	Body & Liner Assembly (Dwg 9340447) Examination for defects	5		3.2	4.4.2.11
	Flash Reducer Loading Assembly (Dwg 9340446) Examination for defects	5		3.2	4.4.2.12

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	CONTINUATION SHEET				
	End Charge & Flash Reducer Loading Assembly (Dwg 9340445) Examination for defects	5		3.2	4.4.2.13
	End Charge and Flash Reducer Loading Assembly and Base Igniter Loading Assembly (Dwg 9340444) Examination for defects	5		3.2	4.4.2.14
	Charge Loading Assembly (Dwg 9340443) Examination for defects	5		3.2	4.4.2.15
	Charge propelling 155MM, M119A2 (Red Bag Chg 7) (in sealed containers) (Dwg and 9333954 and 9333957) Examination for defects (except Proving Ground test)	10		3.2	4.4.2.16/4.4.2.17/ 4.4.2.18/4.4.2.19

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil - Downloaded: 2024-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of propelling charge assemblies shall contain:

a. Propelant from not more than one lot number from one manufacturer.

b. CBI from not more than one interfix number from one manufacturer.

c. Black powder from not more than one interfix number from one manufacturer.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Classifiecation of Defects and Tests Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Strap				9340442
					NEXT HIGHER ASSEMBLY
					9333954
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Width	(a)	0.65%	3.2	Scale
202	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual

NOTES: (a) If a slitting machine is utilized, in lieu of the specified AQL, during the slitting process each roll will be inspected within the first 10 feet and the last 10 feet, i.e., two measurements per roll.

Source: https://assist.dla.mil -- Downloaded: 2020-10-20T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	End, Spot				9340455
					NEXT HIGHER ASSEMBLY
					9340454
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.6
Minor 201	Diameter		0.65%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340450
4.4.2.3	End, Charge				NEXT HIGHER ASSEMBLY 9340445
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Improper cloth			3.1	4.5.6
Minor 201	Diameter		0.65%	3.2	gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil - Downloaded: 2024-10-26 04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340449
4.4.2.4	Laminated Rayon Cloth				NEXT HIGHER ASSEMBLY 9340447
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Established length		0.40%	3.2	gage
102	Established width		0.40%	3.2	gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340448
4.4.2.5	Body				NEXT HIGHER ASSEMBLY 9340447
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.6
Minor 201	Evidence of poor workmanship		0.65%	3.10	visual
202	Marking missing, misleading or unidentifiable		0.65%	3.2	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340456
4.4.2.6	End, Base Igniter				NEXT HIGHER ASSEMBLY 9340452/9340453
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cloth improper			3.1	4.5.6
102	Diameter of base igniter end		0.40%	3.2	gage
Minor					
201	Marking missing, misleading, or unidentifiable		0.65%	3.2	visual
202	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340454
4.4.2.7	Spot Assembly				NEXT HIGHER ASSEMBLY 9340453
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of seam circle		0.40%	3.2	gage
102	Seam incompletely stitched		0.40%	3.2	visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340453
4.4.2.8	End and Spot Assembly				NEXT HIGHER ASSEMBLY 9340452
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
201	Length of bar tack		0.40%	3.2	gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Base Igniter Assembly				9340452
					NEXT HIGHER ASSEMBLY
					9340451
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Diameter of seam circle		0.40%	3.2	gage
Minor 201	Evidence of poor workmanship		0.65%	3.10	visual
202	Marking visible		0.65%	3.2	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340451
4.4.2.10	Base Igniter Loading Assembly				NEXT HIGHER ASSEMBLY 9340444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly incorrect or damaged to the extent that CBI or black powder can escape (including any seams or openings incompletely stitched or broken threads)		100%	3.2	visual/manual
Major					
101	Weight of black powder		100%	3.2	4.5.1
102	Weight of CBI		100%	3.2	4.5.2
103	Any seam improper		0.40%	3.2	visual/gage
104	Moisture content of back powder	(a)		3.3	4.5.5
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES: (a) See para 4.4.3.1

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340447
4.4.2.11	Body and Liner assembly				NEXT HIGHER ASSEMBLY 9340446
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Laminated rayon cloth improper, improperly assembled, missing or improperly located		0.40%	3.2	visual/gage
102	Any seam incompletely stitched		0.40%	3.2	visual/gage
103	Body seam improperly assembled		0.40%	3.2	visual/gage
104	Width of flash reducer pockets		0.40%	3.2	gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340446
4.4.2.12	Flash Reducer Loading Assembly				NEXT HIGHER ASSEMBLY 9340445
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any assembly damaged or improper to the extent potassium sulfate can escape		100%	3.2	visual/manual
102	Total weight of potassium sulfate		0.40%	3.2	balance
103	Any seam incompletely stitched		0.40'%	3.2	visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	End, Charge and Flash Reducer Loading Assembly				9340445
					NEXT HIGHER ASSEMBLY
					9340444
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Any seam incompletely stitched		0.40%	3.2	visual
Minor 201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2024-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340444
4.4.2.14	End Charge and Flash Reducer Loading Assembly and Base Igniter Loading Assembly				NEXT HIGHER ASSEMBLY 9340443
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Seam improper		0.40%	3.2	visual/gage
Minor 201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340443
4.4.2.15	Charge Loading Assembly				NEXT HIGHER ASSEMBLY 9333954
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly incorrect or damaged to the extent that CBI or black powder can escape (including any seams or openings incompletely stitched or broken threads)		100%	3.2	visual/manual
Major					
101	Weight of propellant		100%	3.4.1	4.5.3
102	Any seam improper		0.40%	3.2	visual/scale
103	Charge assembly height		100%	3.8	test
104	Charge diameter (maximum)		100%	3.5.1	gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	visual
202	Marking visible along side of charge		0.65%	3.2	visual

NOTES:

Downloaded from https://assist.dla.mil - Downloaded: 2020-10-27T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Charge, Propelling, 155MM; M119A2 (Red Bag Chg 7)	1 OF 1	9333954
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proving Ground Pressure	(a)		3.6	4.5.7
Major					
101	Dimensional gage failure (maximum length)		0.40%	3.5.1	gage
102	Charge damaged to the extent that propellant or potassium sulfate can escape (includes any seam or openings incompletely stitched and broken threads)		100%	3.2	visual
103	Any component missing or improperly assembled		100%	3.2	visual

NOTES:

(a) Pressure sampling will consist of monitoring pressure on all Proving Ground firings.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9333954
4.4.2.16	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Proving Ground				
104	Muzzle velocity and muzzle velocity standard deviation	15		3.6	4.5.7
Minor 201	Evidence of poor workmanship		0.65%	3.10	visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2020-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Container, Prior to Insertion of Charge				9333957
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Foreign matter or rust in container		100%	3.2	visual
Minor 201	Damaged container (to include dents protruding greater than 1/4" into container)		0.65%	3.2	visual-gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Container, Prior to Sealing				9333957
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing or improper		100%	3.2	visual
102	Charge cannot be removed by hand	(a)		3.2	4.5.8
103	Dimension "X"		0.40%	3.2	gage
Minor	None defined				

NOTES:

(a) One sample from every eight hours production

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63405 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Container, Sealed				9333957
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Test plug missing		100%	3.2	visual
102	DOD symbol missing, misleading or unidentifiable (if applicable)		0.40%	3.2	visual
103	Test plug assembly torque		0.40%	3.7	test
104	Air pressure test		100%	3.2	4.5.4
Minor					
201	Test plug not greased (to be checked prior to reassembly)		0.65%	3.2	visual
202	Marking missing, misleading, or unidentifiable		0.65%	3.2	visual
203	Cover incompletely engaged		0.65%	3.2	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Moisture Content of Black Powder - At least one sample of black powder from each eight hours loading of black powder spot assemblies will be tested for moisture content.

4.4.4 Inspection Equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 of this specification.

4.5 Test Methods and Procedures

4.5.1 Weight of Black Powder - The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the black powder will be inspected 100 percent. The weight of the black powder in the spot will be determined by subtracting an average weight of the empty assembly (no CBI or black powder) or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.2 Weight of CBI - The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the CBI will be inspected 100 percent. The weight of the CBI in the basepad will be determined by subtracting an average weight of the empty assembly (no CBI) or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.3 Propellant Weight - (see 3.4) - The inspections are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.4. The charge loading assemblies will be inspected 100 percent. The weight of the propellant will be determined by subtracting an average or actual weight of the flash reducer loading assembly and end charge assembly and the nominal/actual weight of CBI and Black Powder from the loaded increment weight.

4.5.4 Air Pressure Test - Each container shall be tested as specified on the appropriate drawing. A pressure drop equivalent to or greater than the applicable requirement shall be cause for rejection and removal from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.5.5 Moisture Content of Black Powder - The moisture content of the black powder will be determined in accordance with MIL-P-223. If any sample fails to meet the specified requirement, the eight hours production represented by the sample shall be rejected.

4.5.6 Cloth - At the time cloth is introduced to the sewing and/or cutting operation, an identification will be made for each roll to verify that proper material is used. Any cloth failing to be identified as proper material in accordance with applicable drawing and specifications shall be removed from the sewing and/or cutting operation.

4.5.7 Proving Ground (see 3.6) - This test shall be conducted at a Government Proving Ground in accordance with the applicable acceptance test procedure using M107 projectiles inert loaded within 0.01 pound of an as fired weight of 95.0 pounds using Primer, M82 and fired in cannon M185 with 50 percent (minimum) remaining life. Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature (as indicated by thermocoupled control charges or liquid crystal gages). The following tolerances shall apply to all 145°F and 70°F conditioning: 145°F, from 140°F to 145°F; 70°F, from 68°F to 72°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each charge.

4.5.7.1 Muzzle Velocity and Muzzle Velocity Standard Deviation - Determination of acceptance will be in accordance with MIL-STD-414, Section B, Table B3, AQL of 0.65%. All muzzle velocity and muzzle velocity standard deviation samples will be ballistic tested at 70°F. Fifteen (15) charges will be fired for muzzle velocity and muzzle velocity standard deviation. If the sample is determined unacceptable the lot will be rejected. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.76 exceeds the applicable lot requirement, the lot shall be rejected. Fifteen (15) calibration charges will be fired alternately with the test samples and the muzzle velocity and pressure of all test charges will be corrected to standard conditions.

4.5.7.2 Individual Maximum Pressure (see 3.6) - If any individual corrected pressure observed during firings at 70°F exceeds 37,000 psi, the lot shall be tested at +145°F. For +145°F firings, ten charges will be conditioned to 145°F. Five calibration charges will be fired intermixed with the test samples. If any individual corrected pressure at 145°F exceeds 41,000 psi, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.5.8 Charge Removal From Container - Each sample shall be tested for removal of the propelling charge from the container by hand. The charge may be pulled out by the straps or the container may be inverted and the charge either pulled out or slid out. Any charge that cannot be removed from the container using the above method shall be cause for rejection of the production represented by the sample.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking - The propelling charges shall be packed, sealed, tested and marked in accordance with the specified requirements of drawing 9333957.

6. NOTES

6.1 Intended Use - The propelling charges covered by this specification are intended for use in the 155MM Howitzers, M198 and M109A1/A2/A3.

6.2 Ordering Data - See MIL-A-48078.

6.3 Submission of Designs for Approval - See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Visual Examination Qualification - When a compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.5 Inspection Lot Size - It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions for this specification are maintained.

6.6 Proving Ground - A copy of all proving ground data should be submitted to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R and DRDAR-LCA-G, Dover, New Jersey 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

6.7 Proving Ground Summary

Temp	Sample Size	No of Ref Rds	Velocity	Max Indiv Press (psi)
70°F	15	15	*	*
145°F	10	5	Info	41,000

* See 4.5.7 and 3.6

6.8 Data Cards - A copy of all data cards should be submitted to Commnder, US Army Armament Research and Development Comand, ATTN: DRDAR-QAR-R, Dover, New Jersey 07801.

6.9 Assembly Torque - The assembly torque of the test plug will apply at time of loading only due to the relaxation of the gasket material.

6.10 Muzzle velocity and muzzle velocity standard deviation. This requirement is based on obtaining a nominal velocity of 2245 fps with the M549 RA projectile family in an M199 cannon at zone 7.

6.11 Individual maximum pressure. Individual maximum pressure specified in 3.6 is based on firing an M107 projectile loaded to an as fired weight of 95.0 lbs. Any change in weight or projectile will require an adjustment of this pressure requirement.

Custodian
Army-AR

Preparing Activity
Army-AR

Project Number: 1320-A523

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63405 Charge, Propelling, 155MM M119A2 LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.